The Python script(program) is mainly used to read the C++ file along with the type of mode like 'i'/'o'. 'getopt()' Parses(splits) each value of the command line and passes the options(values) as list to 'opt' and since no error 'args' generates empty list[]. Using 'for loop' the tuple in the list is unpacked - 'o' stores the mode and 'a' stores the name along with the path of the c++ file.

The variable 'inp_file' store the c++ file along with its extension and 'exe_file' stores the executable file with .exe extension. '+' usd in this program helps to concatenate the file name with the extensions. 'os.system()' along with 'g++' compiles the inp_file. Mode 'o' sends the executable file to 'exe_file'.

'__name__' variable directs the program to start from the beginning of the Python script(zero'th line) The "main()" definition does the Parsing and calling the run(). The "run()" invoke the "g++" compiler and creates the exe file. The system() of "os" module executes the .exe file and the desired output will be displayed on the output screen. The file extensions are added by the Python script so it is even possible to execute C programs.

14.8 How Python is handling the errors in C++

Python not only execute the successful C++ program, it also helps to display even errors if any in C++ statement during compilation. For example in the following C++ program an error is there. Let us see what happens when you compile through Python.

Example 14.8.1

```
// C++ program to print the message Hello
//Now select File→New in Notepad and type the C++ program
#include<iostream>
using namespace std;
int main()
{
      std::cout<<"hello"
      return 0;
}
// Save this file as hello.cpp
```

```
# Now select File→New in Notepad and type the Python program as main.py
# Program that compiles and executes a .cpp file
# Python main.py -i  hello
import sys, os, getopt
def main(argv):
   opts, args = getopt.getopt(argv, "i:")
   for o, a in opts:
      if o in "-i":
         run(a)
```


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